05506026 Digital Image Processing (Java)

Lab 4

Objectives

1. Understand how to smooth the image using averaging filter

Averaging filter

Averaging filter is a smooth spatial linear filtering technique to smooth the image. Its purposes are to blur the image and reduce the noise. It can be done by perform average mask on the image.

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

The code to perform averaging filter is as below.

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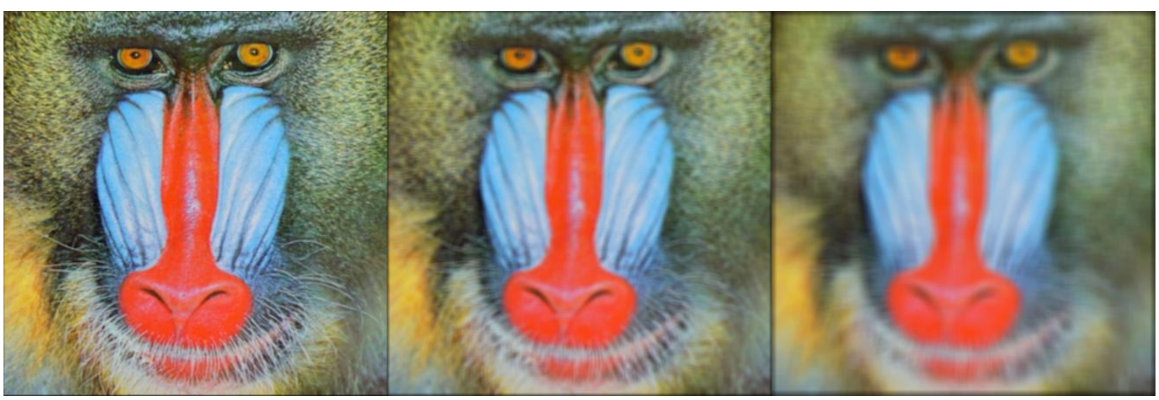
```java
ImageManager.java
class ImageManager
{
    ...
    public void averagingFilter(int size)

    {
        if (img == null)    return;

        if (size % 2 == 0)
        {
            System.out.println("Size Invalid: must be odd number!");
            return;
        }

        BufferedImage tempBuf = new BufferedImage(width, height, img.getType());

        for (int y = 0; y < height; y++)
        {
            for (int x = 0; x < width; x++)
            {

                int sumRed = 0, sumGreen = 0, sumBlue = 0;

                for (int i = y - size/2; i <= y + size/2; i++)
                {
                    for (int j = x - size/2; j <= x + size/2; j++)
                    {
                        if (i >= 0 && i < height && j >= 0 && j < width)
                        {
                            int color = img.getRGB(j, i);
                            int r = (color >> 16) & 0xff;
                            int g = (color >> 8) & 0xff;
                            int b = color & 0xff;

                            sumRed += r;
                            sumGreen += g;
                            sumBlue += b;
                        }
                    }
                }

                sumRed /= (size*size);
                sumRed = sumRed > 255? 255: sumRed;
                sumRed = sumRed < 0? 0: sumRed;
```

```
                sumGreen /= (size*size);
                sumGreen = sumGreen > 255? 255: sumGreen;
                sumGreen = sumGreen < 0? 0: sumGreen;

                sumBlue /= (size*size);
                sumBlue = sumBlue > 255? 255: sumBlue;
                sumBlue = sumBlue < 0? 0: sumBlue;

                int newColor = (sumRed << 16) | (sumGreen << 8) | sumBlue;

                tempBuf.setRGB(x, y, newColor);
            }
        }

        for (int y = 0; y < height; y++)
        {
            for (int x = 0; x < width; x++)
            {
                img.setRGB(x, y, tempBuf.getRGB(x, y));
            }
        }
    }
    ...
}
```

Below is the averaging filter of 3 x 3, 7 x 7, and 15 x 15 on Mandril.bmp respectively.



Quest 006

Implement the method of **median nonlinear filtering** and perform the mask of 3 x 3, 7 x 7, 15 x 15

The player who finishes this task will be awarded **20 EXP** and **2 Moves**.

Deadline: 7 September 2025

Quest 007

Implement the method of using 3 x 3 averaging filter to perform unsharp masking (k = 1)

The player who finishes this task will be awarded **15 EXP** and **1 Move**.

Deadline: 7 September 2025